

AGAI: A Classifier-Anchored, Detector-Grounded Pipeline for Strawberry Disease Diagnosis and Constrained Report Generation

A Reproducibility-Focused Technical Report

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Abstract

We present AGAI, a three-stage pipeline for strawberry disease diagnosis that decouples categorical prediction from natural-language explanation. The pipeline combines (i) a ResNet-50 image classifier [1] that produces a calibrated diagnosis over seven disease states, (ii) an RF-DETR Small detector [2, 3] that grounds the language stage in the plant parts that are actually visible, and (iii) a LoRA-adapted MiniGPT-v2 [4, 5] report writer that explains the classifier diagnosis without re-predicting it. This separation is motivated by the observation that, in agricultural decision support, a confident but unsupported recommendation is at least as harmful as an incorrect class label.

We describe the current dataset state and evaluation protocol. The active evaluation split is a balanced seven-class holdout containing 10 images per class ($n=70$). The compact training root holds 1,506 source images; the augmented training root, rebuilt from the compact root and balanced via deterministic photometric and geometric transforms, holds 2,424 images, with a minimum class floor of 300 for previously underrepresented categories. Holdout exclusion is enforced by physical relocation rather than logical exclusion lists, which closes a class of leakage failures introduced by default directory-tree loaders.

A ResNet-50 checkpoint retrained on the rebuilt augmented root, calibrated by post-hoc temperature scaling [6] ($T = 0.7513$), and deployed with two-view horizontal-flip test-time augmentation, attains 88.57% top-1 accuracy, 92.86% top-2 accuracy, and an expected calibration error of 8.56% on the active holdout. Per-class top-1 accuracy is 100.0% for frost, healthy, and root rot; 90.0% for gray mold; 80.0% for drought and white mold; and 70.0% for overwatered. The RF-DETR Small grounding detector, trained on 753 annotated images with 19,487 instances across six plant parts, reaches a validation mAP₅₀ of 0.772 and mAP_{50:95} of 0.598 (best EMA mAP₅₀ of 0.772 at epoch 93).

At runtime, the pipeline applies per-class confidence thresholding for diagnosis acceptance, injects RF-DETR detections into the report-writer prompt as a visibility constraint, and optionally augments the prompt with retrieval from a curated disease knowledge graph [7]. Stage-2 supervision for MiniGPT-v2 is generated offline under a strict dual-report policy in which an alternative-aware report is emitted only for semantically plausible disease pairs whose top-1/top-2 probability margin lies below a pre-specified threshold. A focused pilot on 20 holdout

images produced 36 records (20 standard, 14 standard variants, 2 ambiguity-aware) with no fallback generations, supporting the feasibility of the redesign while explicitly deferring full-scale regeneration, MiniGPT-v2 retraining, and holdout explanation evaluation as future work.

1 Introduction

Automated plant disease diagnosis exhibits two distinct failure modes that require separate treatment. The first is categorical misclassification, in which a model assigns an incorrect disease label to an input image. The second, which we refer to as *report failure*, occurs when a generative model produces a fluent, detailed explanation that is not in fact supported by the image content. Recent work on multimodal hallucination has documented this failure mode in vision-language systems across general domains [8]. In agricultural decision support, where the consumer of the output is a grower making management decisions, the second failure mode is at least as costly as the first: a confident but unfounded treatment recommendation can damage a crop more reliably than an honest expression of uncertainty.

A single end-to-end vision-language model has structural difficulty addressing both failure modes simultaneously. It must commit to a label, produce a justification for that label, and quantify uncertainty, all from the same generative head. AGAI takes a different approach: we factor the problem into three stages, each with a single responsibility.

1. A calibrated ResNet-50 classifier [1, 6] serves as the sole authority for the categorical diagnosis. The deployed pipeline gates this output by a per-class confidence threshold, demoting low-confidence predictions to an explicit **unknown** state.
2. An RF-DETR Small object detector [2, 3], fine-tuned for strawberry plant parts (*flower, fruit, leaf, root, soil, stem*), produces a visibility constraint that restricts the language model to discussing only those parts that are present in the image.
3. A LoRA-adapted MiniGPT-v2 report writer [4, 5, 9] receives the classifier label and the detector visibility constraint as inputs and is trained to explain rather than re-predict the diagnosis.

This paper documents the system at its current measured state. We make four contributions.

Contribution 1: Reproducible split protocol. We describe and justify a balanced 70-image holdout (10 images per class across seven classes) maintained by physical relocation, an augmented training root rebuilt from a compact source root and balanced to a minimum class floor of 300 by deterministic photometric and geometric augmentation [10], and an external-image audit that adds only conservatively labelled, non-duplicate examples.

Contribution 2: Calibrated classifier evaluation. We retrain a ResNet-50 classifier on the rebuilt augmented root, calibrate it via post-hoc temperature scaling, and report top-1 and top-2 accuracy, expected calibration error [11], per-class confusion, and Wilson 95% confidence intervals [12] for per-class accuracy on the balanced holdout.

Contribution 3: Detector-grounded runtime stack. We formalize the deployed pipeline, including the per-class acceptance gate, the visible-/not-visible-part prompt assembly, optional retrieval from a curated disease knowledge graph, and a heuristic phrase-level hallucination-retry mechanism. Detection performance for the RF-DETR backbone is reported on the held-out validation split using standard COCO metrics [13].

Contribution 4: Constrained dual-report supervision. We describe an offline data-generation policy that produces a primary farmer-facing report for every image and conditionally produces a second, ambiguity-aware report only when the classifier’s top-2 distribution and a pre-specified pair whitelist jointly indicate a semantically plausible alternative. We report results from a controlled 20-image pilot under this policy.

We also identify the experiments that remain open. Full stage-2 regeneration under the constrained policy, MiniGPT-v2 retraining on the regenerated targets, and clean holdout explanation evaluation with and without the RF-DETR grounding constraint are pending. We retain those steps as future work and limit our claims to results obtained from the current artifacts.

2 Related Work

2.1 Image-Based Plant Disease Classification

Image-based plant disease classification has a substantial literature, particularly since the public release of large curated crop image collections such as PlantVillage [14]. Mohanty et al. [15] demonstrated that deep convolutional networks pretrained on ImageNet [16] can be transferred to plant disease classification with strong in-distribution performance, establishing the basic feasibility of smartphone-style diagnosis. Most subsequent work in this tradition terminates at the class label and does not address downstream questions of explanation, uncertainty quantification, or evidence selection. AGAI inherits the classifier-as-anchor design from this body of work but treats explanation as a separate stage with its own constraints.

ResNet-50 [1] remains a strong choice for this setting because residual connections support deep feature extraction without the optimization instability that historically affected very deep CNNs. We do not claim novelty for the classifier architecture; rather, the contribution of the classifier in our pipeline is that it is calibrated, gated, and isolated from the language model.

2.2 Probability Calibration

Modern neural classifiers are commonly miscalibrated, producing overconfident maximum-softmax probabilities. Guo et al. [6] showed that a single-parameter post-hoc temperature scaling correction substantially reduces calibration error without affecting accuracy. We adopt this recipe. Expected calibration error (ECE) [11] is reported for the deployed classifier, and Wilson score intervals [12] are used for per-class accuracy uncertainty due to the small per-class support of the balanced holdout.

2.3 Detection Transformers and Visible-Part Grounding

The DETR family [3] formulates object detection as a direct set prediction problem, eliminating hand-tuned anchors and non-maximum suppression. RF-DETR [2] is a real-time variant in this family and forms the basis of our plant-part detector. In AGAI, the detector serves a non-standard purpose: rather than localize disease symptoms, it produces a binary mask over a fixed six-part vocabulary that is injected into the language model prompt as a visibility constraint. This converts a downstream-only inference task (“what plant parts may be discussed?”) into an upstream perception task with an explicit ground truth.

2.4 Vision-Language Models and Parameter-Efficient Adaptation

The MiniGPT line [4, 17] aligns a vision encoder with a large language backbone via a learned linear projection and lightweight instruction tuning. The closely related LLaVA family [18] demonstrates the broader utility of this class of architectures. AGAI uses MiniGPT-v2 with a Llama 2 chat backbone [9] adapted via low-rank adaptation [5]. The architectural choices are pragmatic; the contribution lies in how the language model is wrapped: as a constrained report writer downstream of an authoritative classifier and detector, rather than as a free diagnostician.

2.5 Retrieval-Augmented Generation and Grounded Output

Retrieval-augmented generation [7] attaches an external non-parametric memory to a generative model so that factual content can be sourced from a curated corpus rather than parameterized weights. We follow this pattern at runtime: a curated disease knowledge graph, populated from cooperative-extension publications, supplies optional treatment-context snippets to the report-writer prompt. The retrieval stage is not the diagnostic authority; it provides citation-backed context for the explanation.

2.6 Hallucination in Multimodal Models

The recent survey of Huang et al. [8] catalogues the landscape of hallucination behaviors in large language models and multimodal systems, including object hallucination, attribute hallucination, and unsupported causal claims. AGAI’s design responds to this literature by reducing the surface area on which the language model can hallucinate: the classifier label is fixed, the visible-part list is fixed, and the generation is restricted to explanation conditioned on both. The constrained dual-report supervision policy described in §4 extends this idea to training-data generation.

2.7 Locality and Inference-Time Cost

Several recent agricultural-AI deployments rely on hosted multimodal APIs at inference time. AGAI deviates from this pattern: the deployed runtime is fully local, with no external model calls during serving. GPT-class models are used only offline during stage-2 supervision generation. Any quality improvement therefore must survive deployment without privileged runtime access to a frontier model.

Table 1: Operational definitions of the seven disease classes, extracted from the runtime disease knowledge graph. The primary visual indicator listed here is the head symptom from the corresponding knowledge-graph entry; classes are not pairwise visually orthogonal, which motivates the constrained dual-report policy of §4.6 for semantically plausible close pairs.

Class	Severity	Primary visual indicator
Drought	moderate	Leaf margins curl inward and become dry, papery, and brittle.
Overwatered	moderate	Persistently wet or saturated growing media; standing water in the root zone.
Root rot	severe	Wilting during midday heat despite adequate soil moisture (water uptake failure).
Frost	variable	Glassy, water-soaked translucent tissue on leaves and petioles immediately after thaw.
Gray mold	severe	Gray, fuzzy sporulation covering fruit surfaces, flowers, or stem lesions (<i>Botrytis cinerea</i>).
White mold	severe	Cottony white mycelium growing on stems, crown base, or fruit (<i>Sclerotinia sclerotiorum</i>).
Healthy	none	Evenly colored dark-green foliage without spots, lesions, or discoloration.

3 Dataset and Split Protocol

3.1 Class Definition and Split Roots

The classifier is trained over seven strawberry health states: *drought*, *frost*, *gray mold*, *healthy*, *overwatered*, *root rot*, and *white mold*. The seven classes are not pairwise visually orthogonal; in particular, the *drought/overwatered* and *overwatered/root rot* pairs share several mid-stage symptoms, and the two mold classes share fruit-surface presentation. Operational definitions and primary visual indicators are summarized in Table 1; the full per-class symptom, cause, and treatment lists are maintained in the disease knowledge graph used by the runtime retrieval module.

The dataset is organized into three on-disk roots, each holding seven class subdirectories:

- the *compact training* root (**train**), which stores the unaugmented source-image pool;
- the *augmented training* root (**train_aug**), the active classifier-training root and stage-2 generation root, derived from **train** via deterministic class-floor balancing;
- the *holdout* root, a physically separate directory tree used for evaluation only.

Per-class counts at the time of writing are reported in Table 2; Figure 1 visualizes the same counts, and Figure 2 shows representative holdout examples.

Table 2: Per-class image counts in the three split roots. Counts reflect the state after the holdout release, augmented-root rebuild, class-floor balancing, and conservative external image audit described in this section.

Class	train	train_aug	holdout
Drought	57	300	10
Frost	109	300	10
Gray mold	602	602	10
Healthy	322	322	10
Overwatered	71	300	10
Root rot	129	300	10
White mold	216	300	10
Total	1506	2424	70

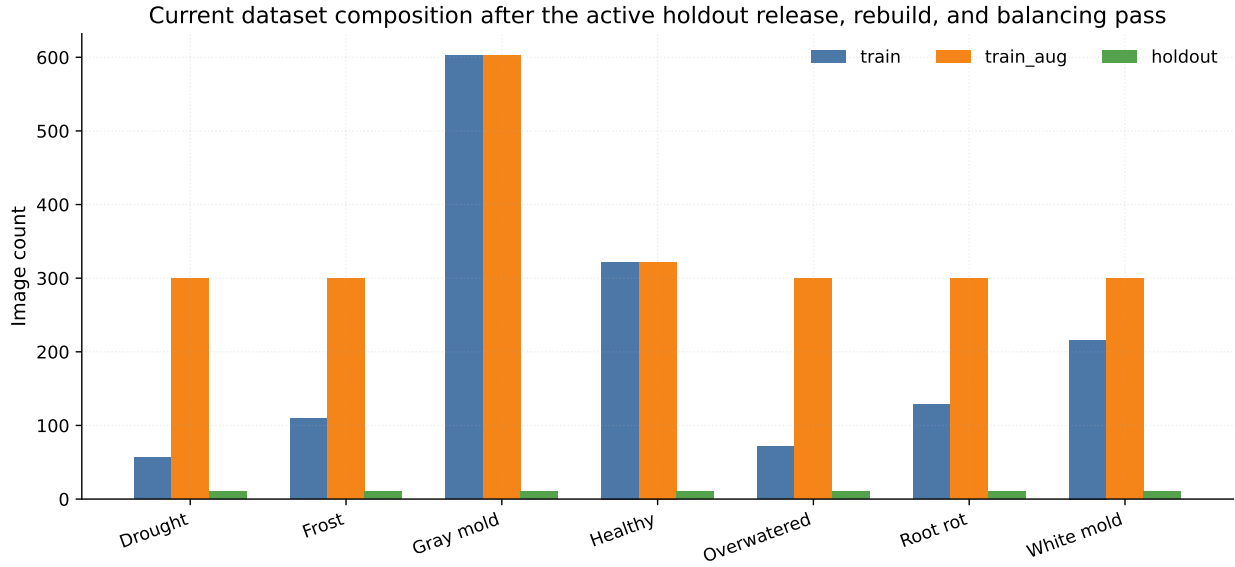


Figure 1: Per-class image counts in the compact training split, the augmented training split, and the active balanced holdout. The augmented split enforces a minimum of 300 images for each underrepresented class.

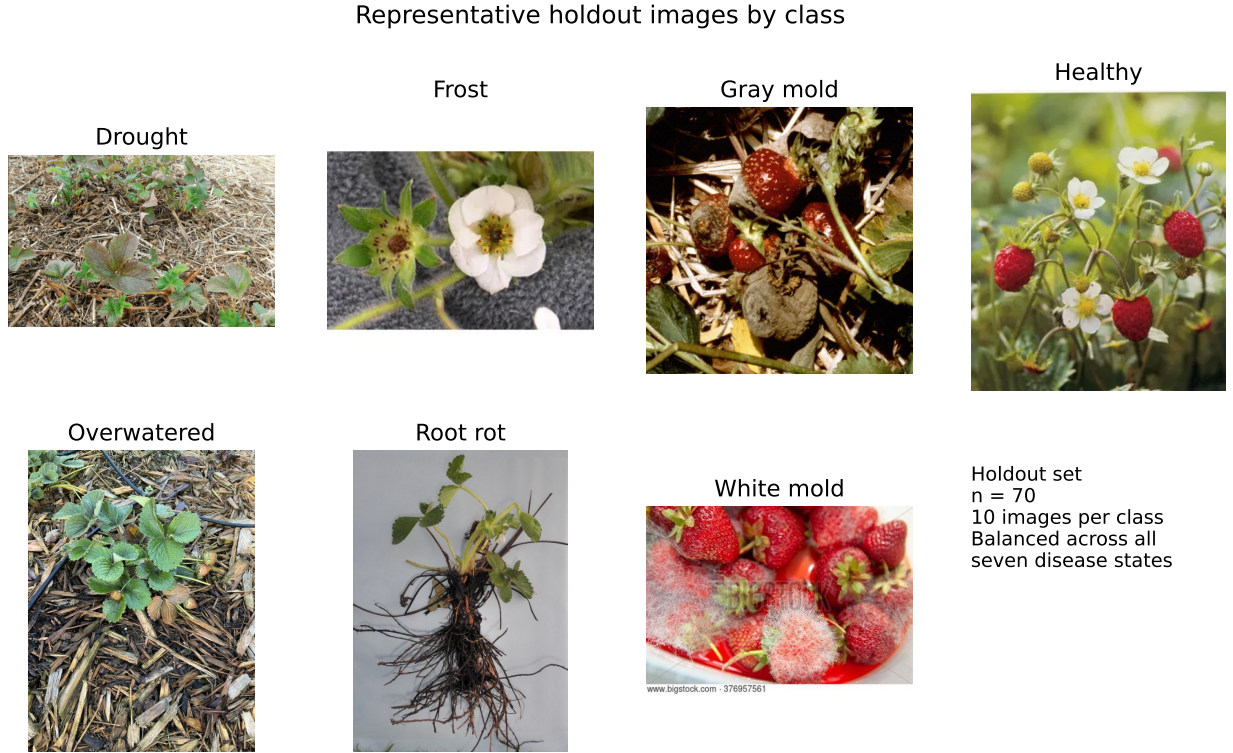


Figure 2: Representative holdout images, one per class. The active holdout contains 10 images per class for a total of $n=70$ evaluation images.

3.2 Holdout Construction by Physical Relocation

The active holdout was constructed by relocating source files out of the training roots, rather than by maintaining a logical exclusion manifest. Logical exclusion is sufficient under disciplined data loading but silently fails when a downstream training script enumerates a directory tree with default settings, which is a common mode in practice. Physical relocation removes the file from the training tree entirely; a default loader cannot re-include it.

The current holdout contains 10 images per class for a total of 70 images. For *gray mold* and *white mold*, the compact and augmented training roots originally matched one-to-one, which simplified holdout selection. For the non-mold classes, the augmented root contained augmentation descendants whose source stems were already present in the training pool. To prevent partial holdout representation in training, each holdout source stem was relocated together with its `_augNNN` descendants.

After an earlier larger-balanced-holdout iteration, 10 images per class were released from the holdout back into the compact training root in order to recover support for underrepresented classes while preserving a balanced evaluation tree. The augmented root was then rebuilt from the compact root rather than patched in place, eliminating any silently surviving balancing artifacts from prior states.

3.3 Class-Floor Balancing

The augmented training root was balanced via a deterministic class-floor augmentation routine that raises any underrepresented class to a minimum of 300 images. The augmentations are deliberately conservative: horizontal flips, small-angle rotations, mild brightness or contrast perturbations, and a flip-plus-rotation composite. This design follows the standard recommendation that low-distortion augmentations preserve the disease appearance in agricultural images more reliably than aggressive synthetic transformations [10]. Classes that already exceeded the floor are not modified.

3.4 Conservative External-Image Audit

In addition to the internal split work, the training state incorporates an earlier human-audited external image audit. After review, six visually defensible non-duplicate images were retained: five *frost* and one *drought*. These images were inserted into the compact training root and given deterministic augmentation descendants in the rebuilt augmented root.

3.5 Effect of the Split Protocol

The current split state addresses two failures of prior iterations. First, the active holdout no longer overrepresents the mold classes relative to the non-mold classes; per-class support is uniform at 10. Second, no holdout source stem (or any of its augmentation descendants) remains in the training roots. Combined with the class-floor balancing, this leaves the augmented training split with no class below the minimum floor of 300 for previously underrepresented categories. We measure all classifier results in §5 on this active holdout.

4 System Architecture

Figure 3 summarizes the pipeline. Solid arrows denote the deployed runtime path; dashed arrows denote the offline stage-2 supervision-generation path used for the dual-report pilot. We describe each stage below.

4.1 Classifier: Backbone, Test-Time Augmentation, and Calibration

The diagnostic classifier is a ResNet-50 [1] with an ImageNet-pretrained backbone [16] and a seven-class linear head. Training uses standard ImageNet normalization, moderate geometric and photometric augmentation, and a 256-pixel evaluation crop. At inference, the classifier averages the softmax probabilities of the original center-cropped image and its horizontal flip. Let $z(x) \in \mathbb{R}^7$ be the logit vector for image x , let $T > 0$ be the temperature learned by post-hoc calibration on the internal validation split, and let $\text{flip}(x)$ denote the horizontal flip of x . The deployed class distribution is

$$\bar{p}(c \mid x) = \frac{1}{2} \left[\text{softmax}\left(\frac{z(x)}{T}\right)_c + \text{softmax}\left(\frac{z(\text{flip}(x))}{T}\right)_c \right], \quad (1)$$

and the predicted label is

$$\hat{y}(x) = \arg \max_{c \in \mathcal{C}} \bar{p}(c \mid x), \quad (2)$$

Current AGAI pipeline. Solid arrows denote deployed runtime flow. Dashed arrows denote the offline data-generation path used for the tightened dual-report pilot. Clean post-carve MiniGPT retraining and holdout explanation evaluation are still pending.

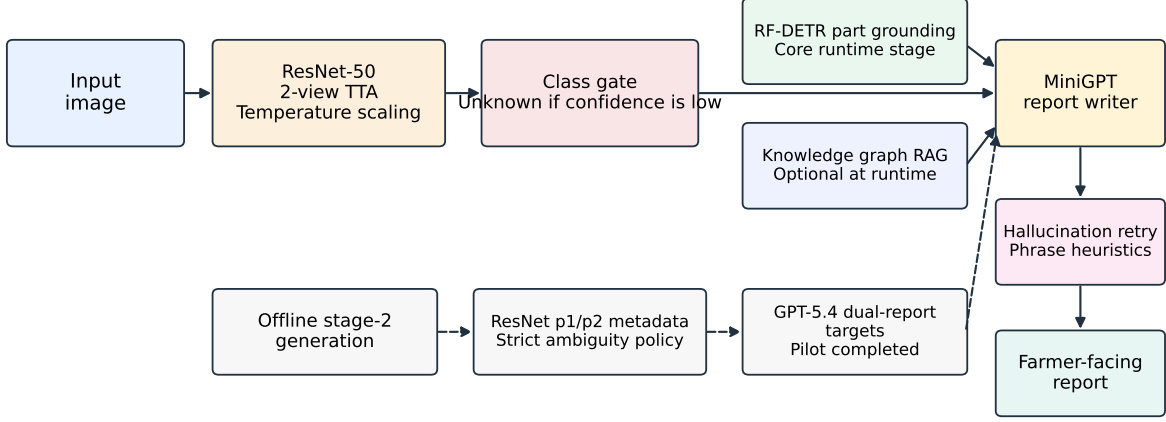


Figure 3: AGAI pipeline. Solid arrows denote the deployed runtime path: ResNet-50 classification with two-view test-time augmentation, per-class threshold gating, RF-DETR part grounding, optional knowledge-graph retrieval, and constrained MiniGPT-v2 report generation. Dashed arrows denote the offline stage-2 supervision-generation path used for the dual-report pilot.

where $\mathcal{C}$ is the seven-class label set. The current deployed checkpoint uses a learned temperature $T = 0.7513$ obtained by minimizing negative log-likelihood on the held-out validation split, following the single-parameter temperature-scaling recipe of Guo et al. [6].

4.2 Runtime Acceptance Gate

Not every classifier output is forwarded to the report writer. The runtime applies a per-class confidence threshold:

$$\text{accept}(x) = \mathbf{1} \left[\bar{p}(\hat{y}(x) \mid x) \geq \tau_{\hat{y}(x)} \right]. \quad (3)$$

The deployed thresholds are

$$\begin{aligned} \tau_{\text{healthy}} &= 0.40, & \tau_{\text{overwatered}} &= 0.60, & \tau_{\text{root rot}} &= 0.65, \\ \tau_{\text{drought}} &= 0.65, & \tau_{\text{frost}} &= 0.70, & \tau_{\text{gray mold}} &= 0.60, & \tau_{\text{white mold}} &= 0.60. \end{aligned} \quad (4)$$

The default for any unmapped class is 0.55. A second guard demotes the prediction to the explicit **unknown** state when the accepted top-1 probability drops below 0.50 after prompt assembly. The runtime additionally records the top-1 and top-2 probabilities p_1, p_2 and the margin $\Delta = p_1 - p_2$ to a structured decision log; a margin threshold of 0.08 is currently logged but not enforced. The classifier accuracy figures reported in §5 are threshold-only and do not depend on margin gating.

4.3 Detector-Grounded Prompt Assembly

Downstream of the classifier, the runtime invokes RF-DETR Small [2, 3] to produce a multi-label visibility vector over the six-part vocabulary $\mathcal{P} = \{\text{flower, fruit, leaf, root, soil, stem}\}$. Per-part

thresholds are tuned to each class:

$$\begin{array}{lll} \theta_{\text{flower}} = 0.50, & \theta_{\text{fruit}} = 0.50, & \theta_{\text{leaf}} = 0.40, \\ \theta_{\text{root}} = 0.30, & \theta_{\text{soil}} = 0.40, & \theta_{\text{stem}} = 0.40. \end{array}$$

Let $V(x) \subseteq \mathcal{P}$ denote the set of parts whose maximum detection confidence exceeds the corresponding threshold. The prompt assembler constructs visible-part and not-visible-part lists, $V(x)$ and $\mathcal{P} \setminus V(x)$, and instructs the language model to describe symptoms only on parts in $V(x)$. A small post-processing heuristic remaps mentions of *leaf* in mold-disease contexts to *fruit* when the surrounding sentence references the calyx, since the green sepals of a strawberry fruit are commonly miscategorized as foliage in free-text descriptions.

4.4 Optional Retrieval and Hallucination Retry

The runtime stack includes two additional support modules:

1. **Knowledge-graph retrieval.** A local disease knowledge base, populated from cooperative-extension sources, exposes a retrieval interface following the pattern of Lewis et al. [7]. When the accepted label is not **unknown**, retrieved snippets are appended to the report-writer prompt as supplemental context. Treatment passages may be conditioned on detected parts (for example, fruit-specific advice when $\textit{fruit} \in V(x)$).
2. **Heuristic hallucination retry.** A phrase-pattern matcher inspects the generated report for a curated list of overclaim triggers (for instance, references to disease names that disagree with the classifier label, or claims about parts in $\mathcal{P} \setminus V(x)$). When a trigger fires, the runtime re-issues the generation. This module is explicitly heuristic and has not yet been calibrated against a false-positive budget.

All three modules execute locally; no external model is queried at inference time.

4.5 Report Writer Configuration

The report-writer backbone is MiniGPT-v2 [4], with a Llama 2 chat backbone [9] and LoRA adaptation [5]. The active training configuration uses 448-pixel image inputs, a maximum text length of 500 tokens, LoRA rank $r = 64$ with $\alpha = 128$ and dropout 0.05, batch size 2 per GPU with 8 gradient-accumulation steps, and 20 epochs with a 10% auto-validation split. These are the active repository settings; we note in §5 that the deployed checkpoint predates the constrained stage-2 redesign.

4.6 Constrained Dual-Report Supervision Policy

Stage-2 supervision targets are generated offline from a frontier model (used only at dataset-construction time). For each input image, the policy emits a primary farmer-facing report. A second report is emitted only if the classifier exposes a *semantically plausible* close alternative, where plausibility is determined by both a margin condition and a curated pair whitelist.

Let $g(x)$ denote the ground-truth folder label, $y_1(x)$ the top-1 classifier label, and $y_2(x)$ the top-2 classifier label. The competing class used by the policy is

$$a(x) = \begin{cases} y_2(x), & y_1(x) = g(x), \\ y_1(x), & y_1(x) \neq g(x). \end{cases} \quad (5)$$

The allowed pair set is

$$\mathcal{A} = \left\{ \begin{array}{l} \{\text{gray mold, white mold}\}, \\ \{\text{drought, overwatered}\}, \\ \{\text{drought, frost}\}, \\ \{\text{healthy, overwatered}\}, \\ \{\text{root rot, overwatered}\}. \end{array} \right\} \quad (6)$$

A second report is labelled *ambiguous* only when one of two conditions holds:

$$\begin{aligned} y_1(x) = g(x), \quad \bar{p}(y_1 | x) - \bar{p}(y_2 | x) \leq 0.10, \\ \bar{p}(y_2 | x) \geq 0.15, \quad \{g(x), y_2(x)\} \in \mathcal{A}, \end{aligned} \quad (7)$$

$$\begin{aligned} y_1(x) \neq g(x), \quad \text{rank}(g(x)) = 2, \\ \bar{p}(y_1 | x) - \bar{p}(g(x) | x) \leq 0.10, \\ \bar{p}(g(x) | x) \geq 0.15, \quad \{g(x), y_1(x)\} \in \mathcal{A}. \end{aligned} \quad (8)$$

When the classifier is correct and the margin is wide, the second report is emitted as a *standard variant*: a paraphrase under the same diagnostic label, used to expand training-target diversity. When the classifier is incorrect and the ground-truth label is not a close rank-2 competitor, no second report is emitted, which prevents classifier failures from being relabelled as ambiguity events. The policy is restricted to dataset construction and does not currently imply any user-facing alternative-presentation behavior at runtime.

5 Experimental Results

We report results in three layers: classifier evaluation on the active holdout (§5.1), grounding-detector performance on its validation split (§5.2), and the constrained dual-report pilot (§5.3). End-to-end explanation evaluation, which requires a MiniGPT-v2 retrain on regenerated stage-2 supervision, remains pending and is discussed explicitly in §5.4.

5.1 Classifier Evaluation on the Balanced Holdout

The deployed ResNet-50 v4 checkpoint was retrained on the rebuilt `train_aug` root described in §3 and evaluated on the active 70-image holdout under the two-view test-time augmentation and post-hoc temperature-scaling protocol of Eq. 1. The aggregate metrics are reported in Table 3.

Table 5: Per-class holdout performance for the deployed ResNet-50 v4 checkpoint. Wilson 95% confidence intervals [12] are reported because the per-class support of $n = 10$ is small. Both top-1 and top-2 accuracy are computed under the two-view TTA protocol.

Class	n	Top-1	Wilson 95% CI	Top-2
Drought	10	80.0%	[49.0, 94.3]	90.0%
Frost	10	100.0%	[72.2, 100.0]	100.0%
Gray mold	10	90.0%	[59.6, 98.2]	100.0%
Healthy	10	100.0%	[72.2, 100.0]	100.0%
Overwatered	10	70.0%	[39.7, 89.2]	80.0%
Root rot	10	100.0%	[72.2, 100.0]	100.0%
White mold	10	80.0%	[49.0, 94.3]	80.0%

Table 3: Aggregate holdout metrics for the deployed ResNet-50 v4 checkpoint on the active balanced 70-image holdout. “Temperature” refers to the value T in Eq. 1.

Metric	Current v4
Holdout samples	70
Top-1 accuracy	88.57%
Top-2 accuracy	92.86%
ECE	8.56%
Temperature	0.7513
Horizontal-flip TTA	yes

The checkpoint attains 88.57% top-1 accuracy, 92.86% top-2 accuracy, and an expected calibration error [11] of 8.56% under the deployed two-view TTA protocol. We emphasize that these metrics are reported on the active dataset state described in §3, which combines a balanced holdout, a rebuilt augmented training root, a class-floor balancing step for previously underrepresented classes, and a conservative external-image audit, with explicit physical exclusion of holdout source stems and their augmentation descendants from the training roots.

To quantify uncertainty on the aggregate metrics under the small holdout size, we report nonparametric percentile bootstrap 95% confidence intervals (10,000 resamples, seed 42) over the per-image correctness indicators in Table 4.

Table 4: Nonparametric bootstrap 95% confidence intervals for the deployed v4 checkpoint on the balanced 70-image holdout.

Aggregate metric	Point estimate	95% bootstrap CI
Top-1 accuracy	88.57%	[80.00, 95.71]
Top-2 accuracy	92.86%	[85.71, 98.57]

Per-class metrics, including Wilson 95% confidence intervals [12], are reported in Table 5.

Frost, healthy, and root rot each reach 100.0% top-1 accuracy on $n = 10$. Gray mold reaches 90.0%, drought and white mold each reach 80.0%, and overwatered reaches 70.0%. Per-class supports are uniform, which permits direct comparison across classes; however, the small support implies wide

confidence intervals on individual rows, which should be considered when interpreting between-class differences.

Figure 4 reports the confusion matrix on the active holdout. The dominant off-diagonal mass falls on the overwatered row, where two examples are misclassified as drought and one as frost, and on the white-mold row, where one example is misclassified as gray mold and one as healthy. Figure 5 reports per-class top-1 accuracy and the reliability diagram for the calibrated v4 checkpoint.

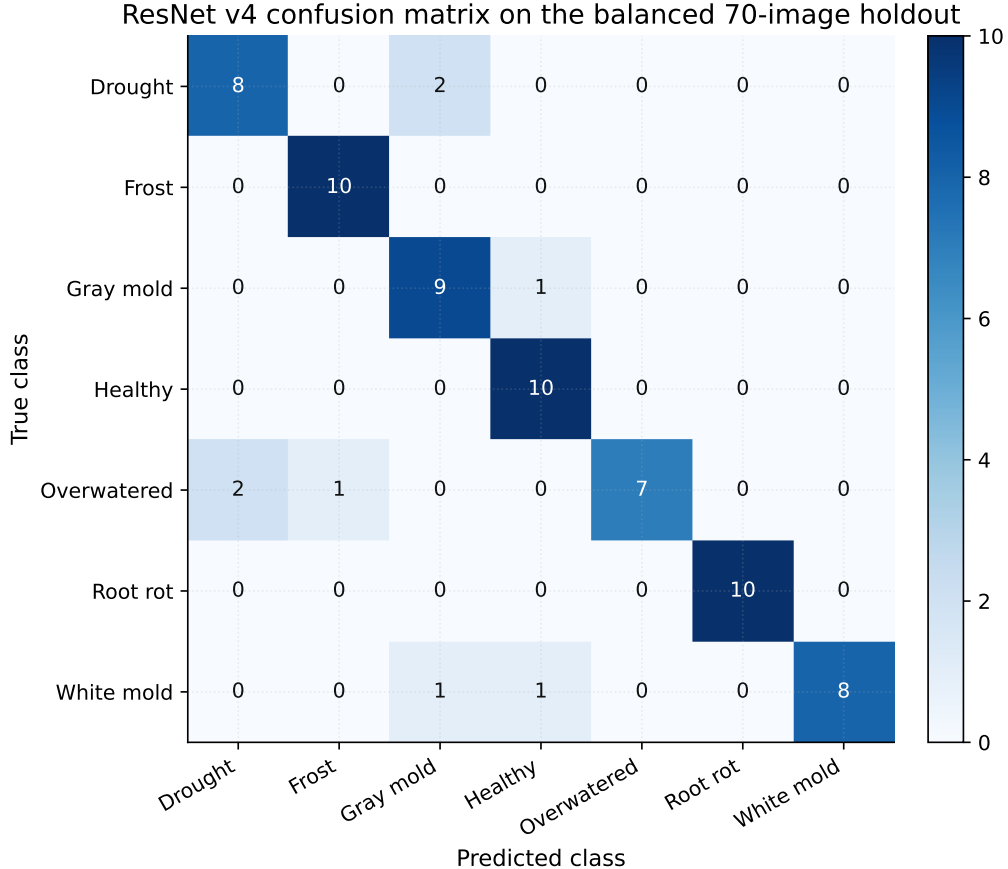


Figure 4: Confusion matrix for the deployed ResNet-50 v4 checkpoint on the balanced 70-image holdout. Rows correspond to ground-truth classes; columns to predicted classes.

The threshold-only acceptance gate of Eq. 3 is the deployed runtime decision rule. We do not report margin-gated metrics in this manuscript because the margin threshold of 0.08 is currently recorded for analysis only (§4.2); margin-rule activation will be informed by the production decision logs once sufficient real-world traffic has accumulated.

Acceptance-gate behavior on the holdout.

To quantify the effect of the runtime decision rule on the same evaluation distribution, we replay the deployed gate on the v4 predictions: for each holdout image, we accept the top-1 label only when its calibrated probability exceeds both the per-class threshold τ_c in Eq. 4 and the post-prompt-assembly floor of 0.50, otherwise we demote the prediction to **unknown**. Table 6 reports per-class acceptance

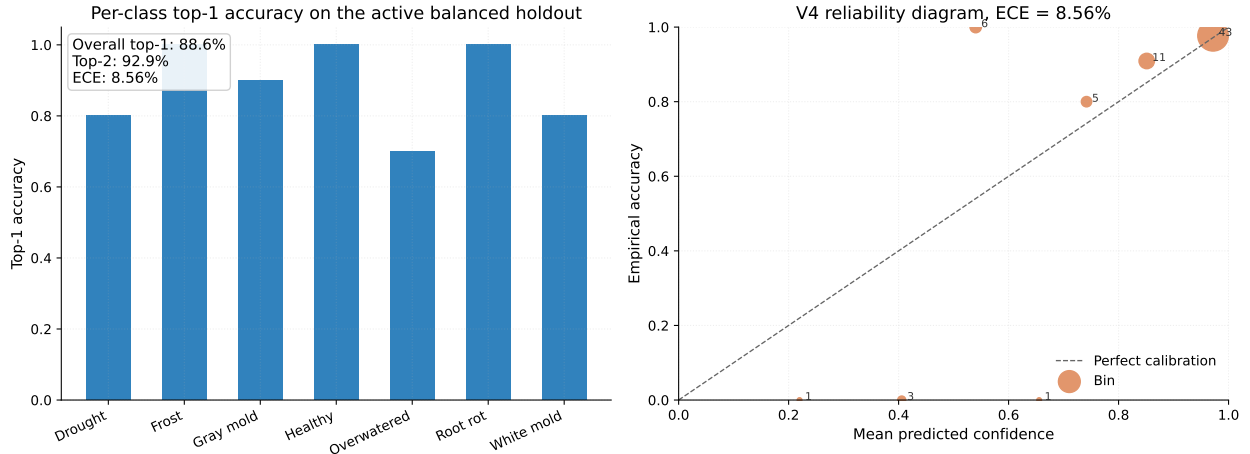


Figure 5: Left: per-class top-1 accuracy on the balanced holdout. Right: reliability diagram for the deployed v4 checkpoint under two-view horizontal-flip TTA. Marker area is proportional to bin support; the diagonal denotes perfect calibration. The expected calibration error is 8.56%.

Table 6: Acceptance-gate behavior on the balanced holdout. “Acceptance rate” is the fraction of class examples whose top-1 probability clears both τ_c and the 0.50 floor; “Acc. | accepted” is the conditional accuracy on the accepted subset. The aggregate row reports a nonparametric bootstrap 95% CI on conditional accuracy.

Class	n	Accepted	Acceptance rate	Acc. accepted
Drought	10	6	60.0%	83.3%
Frost	10	9	90.0%	100.0%
Gray mold	10	8	80.0%	100.0%
Healthy	10	10	100.0%	100.0%
Overwatered	10	9	90.0%	77.8%
Root rot	10	10	100.0%	100.0%
White mold	10	9	90.0%	88.9%
Total	70	61	87.1%	93.4% [86.9, 98.4]

counts, the acceptance rate, and the conditional accuracy on the accepted subset.

The deployed gate accepts 61 of 70 holdout images (87.1%), and the conditional top-1 accuracy on the accepted set rises to 93.4% (95% bootstrap CI: 86.9% – 98.4%) compared with 88.57% on the unfiltered holdout. The remaining 9 images are demoted to **unknown**, which is the operationally desired behavior for images on which the classifier is not confident enough to drive a report.

Threshold sensitivity and risk-coverage behavior.

To characterize the gate beyond a single operating point, we sweep a uniform threshold $\tau \in [0, 0.99]$ over the calibrated top-1 probability and report acceptance rate and conditional accuracy (Figure 6). Ordering predictions by descending p_1 produces the corresponding risk-coverage curve in Figure 7; selective top-1 accuracy at fixed coverage levels is reported in Table 7.

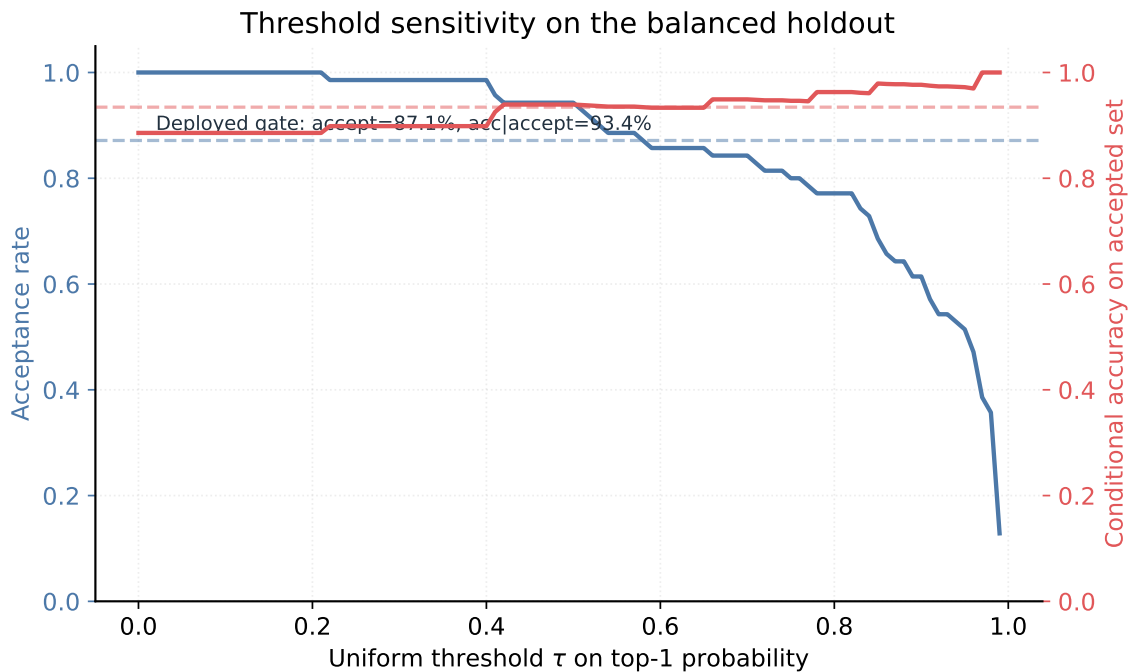


Figure 6: Threshold sensitivity on the balanced holdout under a uniform threshold τ on $\bar{p}(\hat{y} | x)$. The dashed lines mark the deployed (per-class) gate operating point.

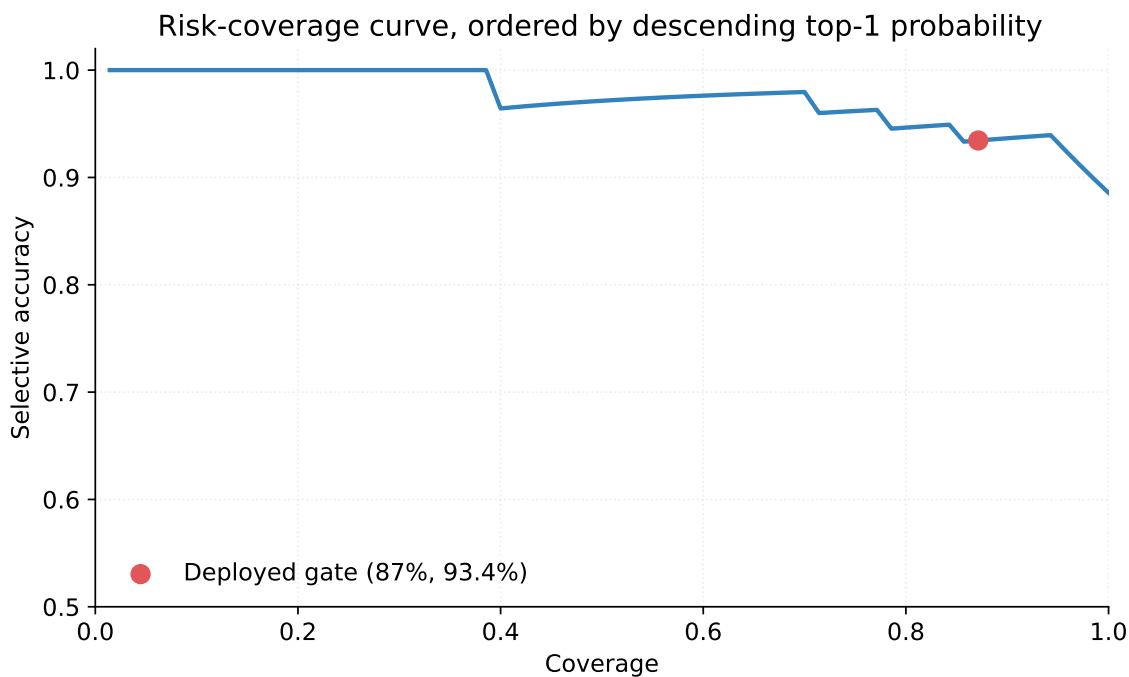


Figure 7: Risk-coverage curve obtained by ordering holdout predictions by descending top-1 probability. The red marker shows the operating point of the deployed per-class gate.

Table 7: Selective top-1 accuracy at fixed coverage levels on the balanced holdout, computed by ordering predictions by descending top-1 probability.

Coverage	Accepted images	Selective top-1 accuracy
50%	35	97.14%
70%	49	97.96%
85%	60	93.33%
100%	70	88.57%

Paired comparison of v3 and v4.

To test whether the v4 retrain produces a statistically distinguishable improvement over the prior v3 checkpoint on the same holdout, we performed a McNemar paired test (Table 8). Of the 70 holdout images, both checkpoints agree on 64 (58 both correct, 6 both incorrect); the v4 retrain is uniquely correct on 4 images and v3 is uniquely correct on 2. The continuity-corrected $\chi^2_{1\text{df}}$ statistic is 0.167 (asymptotic two-sided $p = 0.683$; exact binomial two-sided $p = 0.688$), which we report explicitly: the +2.86 percentage-point improvement in top-1 accuracy from v3 (85.71%) to v4 (88.57%) is not statistically significant at $n = 70$. We retain the v4 checkpoint because of (i) the parallel split hygiene that motivated the retrain (§3) and (ii) the detector and supervision changes that depend on the same post-rebuild artifacts.

Table 8: McNemar paired test comparing v3 and v4 ResNet-50 checkpoints on the shared 70-image holdout.

Quantity	Value
Holdout size n	70
Both correct (n_{11})	58
Only v3 correct ($n_{10} \equiv b$)	2
Only v4 correct ($n_{01} \equiv c$)	4
Both incorrect (n_{00})	6
v3 top-1 accuracy	85.71%
v4 top-1 accuracy	88.57%
McNemar χ^2 (continuity-corrected, 1 df)	0.167
χ^2 asymptotic two-sided p	0.683
Exact binomial two-sided p	0.688

5.2 RF-DETR Grounding-Detector Evaluation

The grounding detector is an RF-DETR Small instance fine-tuned on a COCO-style dataset of 753 training images and 322 validation images, containing 19,487 training and 8,582 validation instance annotations across the six-part vocabulary defined in §4. Training uses 100 epochs at resolution 560 with batch size 8 and gradient accumulation factor 2 (effective batch 16), encoder learning rate 1.5×10^{-4} , decoder learning rate 1×10^{-4} , weight decay 1×10^{-4} , cosine schedule with five warmup epochs, drop-path 0.1, and EMA enabled.

Final-epoch validation metrics are summarized in Table 9. The reported quantities follow the COCO evaluation protocol [13]: average precision at intersection-over-union threshold 0.5 (AP_{50}), AP_{75} ,

the standard $\text{AP}_{50:95}$ average, and recall-style AR. Best EMA mAP_{50} on the validation split is 0.7723 at epoch 93.

Table 9: RF-DETR Small validation metrics at the final training epoch. Class-conditional AP is reported at IoU threshold 0.5; aggregate quantities follow the COCO evaluation protocol [13].

Quantity	Value
mAP_{50}	0.772
$\text{mAP}_{50:95}$	0.598
mAP_{75}	0.644
F_1	0.737
Precision	0.798
Recall	0.687
AP_{50} flower	0.611
AP_{50} fruit	0.697
AP_{50} leaf	0.590
AP_{50} root	0.482
AP_{50} soil	0.710
AP_{50} stem	0.498
EMA mAP_{50} (best, epoch 93)	0.772

The intended runtime use of the detector is constraint generation, not high-fidelity localization. Per-class detection thresholds θ_p given in §4 were tuned to favor recall on parts whose absence is more frequently asserted in free-text reports than presence (e.g., *root*, $\theta_{\text{root}} = 0.30$), and to suppress false positives on the more visually distinct parts (*flower*, *fruit*, $\theta = 0.50$). A formal on/off ablation of the grounding constraint, evaluated against the language stage on the holdout, is identified as future work in §5.4.

Detector behavior on the disease holdout.

The metrics in Table 9 are computed on the detector’s own COCO validation split. To characterize behavior on the deployment distribution, we additionally ran the deployed detector on the 70 disease-holdout images and recorded the per-class rate at which each plant part fires above its runtime threshold θ_p (Table 10; visualized in Figure 8). Median single-image inference latency is 18.4 ms (P95 144.2 ms) on a single RTX 3090. The disease holdout contains no part-level ground truth, so we report visibility-firing rates rather than precision and recall.

Table 10: Per-disease plant-part detection rates of the deployed RF-DETR Small checkpoint on the 70 disease-holdout images, using the runtime thresholds θ_p of §4. The reported quantity is the fraction of class images for which the corresponding part fires at least once above θ_p . “Mean parts” is the average number of distinct visible parts per image.

Disease	n	Flower	Fruit	Leaf	Root	Soil	Stem	Mean parts
Drought	10	0%	60%	100%	0%	40%	90%	2.90
Frost	10	60%	30%	100%	0%	30%	90%	3.10
Gray mold	10	30%	100%	90%	0%	0%	90%	3.10
Healthy	10	50%	100%	100%	0%	40%	80%	3.70
Overwatered	10	0%	10%	90%	10%	80%	70%	2.60
Root rot	10	0%	0%	90%	100%	20%	70%	2.80
White mold	10	0%	100%	100%	0%	0%	20%	2.20
All classes	70	20%	57%	96%	16%	30%	73%	2.91

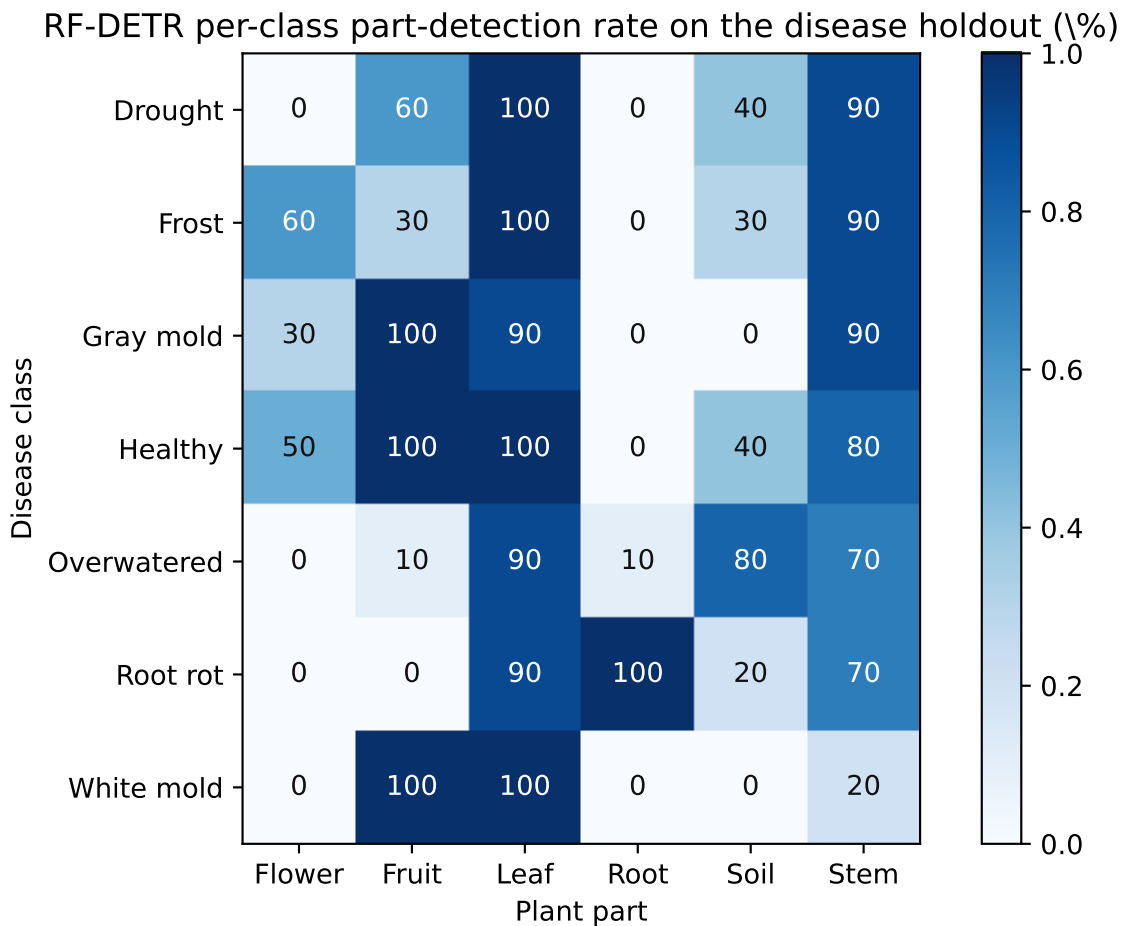


Figure 8: Per-class plant-part detection rate (%) of the deployed RF-DETR Small checkpoint on the disease-holdout images.

The pattern is consistent with the deployment intent of the detector: *leaf* fires on 95.7% of images across classes, the *root rot* row fires *root* on 100% of its images, and *soil* fires preferentially on

overwatered (80%) where saturated substrate is the dominant visual cue. Detection rates for *flower* and *root* are otherwise low because those parts are infrequently visible in the holdout distribution rather than because the detector is failing.

5.3 Constrained Dual-Report Pilot

The dual-report pilot is a controlled evaluation of whether the supervision policy of §4.6 produces useful training targets, rather than a full retraining of the report writer. The selection routine drew 20 holdout images stratified by class and rule type. The selection rule counts were 14 close-or-clear-margin (top-1 matches ground truth) cases of which 2 were close-margin, and 4 top-1-disagrees-with-ground-truth hard cases. Pilot summary statistics are reported in Table 11 and visualized in Figure 9.

Table 11: Statistics for the constrained dual-report pilot (§4.6). Selection rule counts and report-mode counts are reported separately. “Self-reported overclaim risk” is a generator metadata field that records the model’s own coarse self-rating of overclaim risk per record.

Focused tight dual-report pilot statistic	Value
Selected images	20
Generated records	36
Hard-case selections	4
Close-margin selections	2
Standard reports	20
Standard variants	14
Ambiguous reports	2
Self-reported low overclaim risk	28
Self-reported medium overclaim risk	8
Fallback reports	0

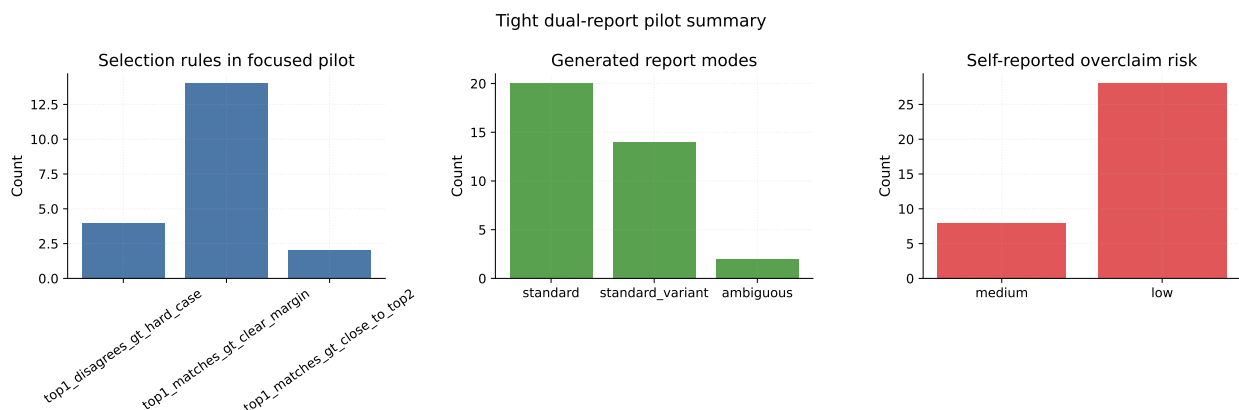


Figure 9: Constrained dual-report pilot summary. After the policy was restricted to the pair whitelist of Eq. 6 and the margin condition of Eq. 7–8, ambiguity reports are rare and concentrated on semantically plausible disease pairs.

The pilot produced 36 records: 20 standard reports, 14 standard variants, and 2 ambiguity-aware

reports. No record required the fallback generation path. The constrained ambiguity policy filters out semantically implausible second-rank competitors that arise from numerical proximity alone; the surviving ambiguity records correspond to disease pairs that are operationally and visually realistic.

The pilot supports three limited claims:

1. the dual-report generation pipeline executes end-to-end without fallback failures;
2. the constrained policy suppresses semantically implausible ambiguity selections; and
3. the resulting target distribution is sufficient to motivate full-scale stage-2 regeneration under the same policy.

The pilot does *not* constitute a measurement of post-retraining explanation quality on the holdout; the deployed MiniGPT-v2 checkpoint predates the constrained policy.

5.4 Pending Experiments

Several experiments are deliberately deferred and are not reported in this manuscript:

1. **Stage-2 regeneration at scale.** The current stage-2 training annotations do not yet encode the constrained policy of §4.6 on the full augmented training set.
2. **MiniGPT-v2 retraining.** The deployed checkpoint predates the constrained policy and the holdout carve. A retrained checkpoint is required before any post-policy explanation claim can be supported.
3. **Holdout explanation evaluation.** No post-regeneration explanation rubric has been applied to the retrained checkpoint.
4. **Grounding ablation.** An on/off ablation of the RF-DETR visibility constraint, evaluated against the language stage on the holdout, is planned but not executed.

These items separate a current-state technical report (this paper) from a finished end-to-end benchmark.

6 Discussion

6.1 Asymmetry Between Classifier and Language Claims

The classifier and the language stages of the pipeline are at different levels of empirical maturity, and the manuscript reflects that asymmetry. On the classifier side, the methodology is now substantially more rigorous than in earlier development phases: the active holdout is physically separated from the training roots, per-class support is balanced at $n = 10$, the augmented training root has been rebuilt from the compact root with a class-floor balancing pass, and a fresh v4 checkpoint has been retrained on the rebuilt root and evaluated on the balanced holdout under calibrated two-view TTA. The classifier results in §5.1 are therefore reproducible from the recorded artifacts.

On the language side, the picture is more partial. The runtime integration of MiniGPT-v2, the RF-DETR visibility constraint, the optional retrieval module, and the heuristic hallucination retry are deployed and exercised on a regular basis. The constrained dual-report supervision policy has been validated by the focused pilot described in §5.3. However, the deployed report-writer checkpoint predates the constrained stage-2 redesign and the balanced holdout carve, which means any explanation-quality measurement on the current holdout would be confounded by training-time exposure. Accordingly, we restrict the language claims of this manuscript to (i) a description of the runtime integration and (ii) a feasibility result for the constrained supervision policy. We do not assert post-policy explanation quality.

6.2 Decoupling Classifier Quality from Deployed Diagnosis Rate

The threshold gate of Eq. 3 introduces a deliberate gap between classifier accuracy on the holdout and the deployed diagnosis-acceptance rate. A correct prediction can be demoted to **unknown** when its top-1 probability falls below $\tau_{\hat{y}}$, and a marginal correct prediction can be subsequently demoted by the post-prompt-assembly 0.50 guard. We do not retune the thresholds in this manuscript because the deployed runtime now logs the top-1 and top-2 probabilities to a structured decision log, and we prefer to retune from real decision data rather than from holdout evaluation alone. The current manuscript therefore reports classifier accuracy as a property of the model, not as the deployed user-facing diagnosis rate.

6.3 Upstream–Downstream Coupling for Ambiguity

The constrained dual-report policy underscores a more general property of the system: usefulness of the language-stage uncertainty behavior is not determined by the language model alone. A naive “top-2 implies plausible alternative” rule would surface numerically close but semantically implausible competitors (for example, frost as the rank-2 competitor for healthy on a uniformly green plant), and these would become legitimate-looking ambiguity reports. The policy becomes defensible only after both a margin condition (Eq. 7–8) and a curated pair whitelist (Eq. 6) are applied. This is a concrete instance of a more general principle for systems in which a fluent generator follows a structured predictor: the generator’s uncertainty expression is only as well-formed as the upstream filter that produces its inputs.

6.4 Detector Use as Constraint, Not Localization

The deployed RF-DETR Small detector reaches a validation mAP₅₀ of 0.772 and mAP_{50:95} of 0.598. These numbers are competitive for a six-class plant-part vocabulary at the training data scale described in §5.2, but they are also not the full criterion for the detector’s intended runtime role. The detector is deployed as a binary visibility oracle, not as a fine-grained localizer: the report writer consumes the set $V(x)$ of parts for which any detection clears the per-part threshold θ_p , not the bounding boxes themselves. We tune θ_p in favor of recall on parts whose absence is more frequently asserted than their presence (in particular *root*). A formal ablation of the visibility constraint against an unconstrained baseline is identified as future work.

6.5 Limits of Local Inference

A persistent design constraint is that the deployed runtime must be fully local: no frontier-model API is queried at inference time. Any quality improvement therefore has to survive deployment without privileged runtime access to a frontier model. In practice, this places the burden on stage-2 supervision quality rather than on online distillation or runtime auxiliary scoring. The constrained dual-report policy is the principal tool we use to translate frontier-model quality into local-model quality through the offline supervision channel.

6.6 Threats to Validity

We note the following threats to validity.

Small per-class support. With $n = 10$ per class, both Wilson and bootstrap intervals on per-class top-1 accuracy are wide (Table 5). Observed differences between classes should be treated as suggestive rather than significant; the aggregate bootstrap interval in Table 4 (80.0–95.7% top-1) is the appropriate level at which to compare the deployed checkpoint to alternatives.

Single-distribution evaluation. The active holdout reflects a single-cultivar, single-region image distribution. The manuscript does not include cross-cultivar or cross-region generalization, and we do not claim it. This is a deliberate scope restriction: external evaluation is identified as future work alongside the experiments listed in §5.4.

Calibration data overlap. The temperature parameter T in Eq. 1 is fit on the internal validation split of the augmented training root, and ECE is reported on the disjoint physical holdout. However, both splits are drawn from the same source distribution, so the reported ECE is an in-distribution calibration estimate rather than a cross-distribution estimate.

Detector measurement is firing-rate, not precision/recall. The RF-DETR mAP metrics in Table 9 are computed on the detector’s COCO validation split. We additionally report visibility firing rates of the deployed detector on the disease-holdout distribution (Table 10), but those firing rates do not constitute precision/recall numbers because the disease holdout has no plant-part ground truth. Per-disease firing rates are consistent with what the deployment intent of the detector requires (e.g., 100% *root* firing on *root rot*; 80% *soil* firing on *overwatered*), but a like-for-like detection benchmark on the disease distribution would require additional plant-part annotation of the holdout images.

v3-to-v4 retrain not significant at $n = 70$. The McNemar paired test in Table 8 reports a two-sided exact $p = 0.688$ for the comparison of v3 (85.71%) and v4 (88.57%) on the shared 70-image holdout. The retrain is therefore motivated by split-hygiene reasons (§3) rather than by a statistically significant accuracy improvement at this sample size.

Pilot scope. The dual-report pilot is a 20-image feasibility result. It supports claims about pipeline executability and policy filtering, but does not constitute a measurement of post-retraining

explanation quality. The pending experiments in §5.4 are required before stronger end-to-end claims can be supported.

Heuristic hallucination retry. The phrase-trigger retry module is not calibrated against a false-positive budget; we do not characterize its precision or recall on any test distribution. It is described in §4 as part of the deployed runtime, but no claim about its effect on report quality is made.

6.7 Summary

The deployed AGAI pipeline supports a stronger classifier claim than language claim, and the manuscript reports them at the level of evidence they currently support: the classifier and detector results are calibrated and measured, while the language stage is described and shown feasible under the constrained supervision policy without yet being benchmarked end-to-end.

7 Conclusion

We described AGAI, a strawberry disease diagnostic pipeline that factors the task into a calibrated classifier, a part-level visibility detector, and a constrained report writer. On the active balanced seven-class holdout, the deployed ResNet-50 v4 checkpoint attains 88.57% top-1 accuracy, 92.86% top-2 accuracy, and 8.56% expected calibration error under two-view test-time augmentation and post-hoc temperature scaling, with 100.0% per-class top-1 accuracy on frost, healthy, and root rot. The RF-DETR Small grounding detector reaches a validation mAP₅₀ of 0.772 and mAP_{50:95} of 0.598, supporting its runtime use as a binary visibility oracle for the language stage.

We further described and exercised a constrained dual-report supervision policy that produces a primary report for every image and an ambiguity-aware second report only when both a probability-margin condition and a curated semantic-pair whitelist are satisfied. A focused 20-image pilot under this policy produced 36 records (20 standard, 14 standard variants, 2 ambiguity) with no fallback generations, supporting the feasibility of the redesign as a target-distribution change before full-scale regeneration.

Three classes of experiments remain open and are explicitly deferred: full-scale regeneration of stage-2 supervision under the constrained policy, MiniGPT-v2 retraining on the regenerated targets, and clean holdout explanation evaluation with and without the visibility constraint. We frame those as future work rather than as missing results, and we limit the present manuscript’s claims to the classifier, detector, and pilot evidence currently in hand. Within those limits, the manuscript documents a reproducible split protocol, a calibrated classifier evaluation, a measured detector backbone, and a controlled feasibility study of the constrained supervision policy.

A Appendix: Configuration and Reproducibility Notes

A.1 Project Status Summary

Table 12 summarizes the state of the components referenced in the main text. The classifier and detector evidence are sufficient to support the results in §5; the language-stage components remain

in transition.

Table 12: Status summary for the components referenced in this manuscript.

Component	Status	Notes
Balanced holdout with provenance	Done	The active holdout now contains 10 images for each of the seven classes
Compact-train release and train_aug rebuild	Done	Ten images per class were released from hold-out back to train and the augmented root was rebuilt from scratch
Augmented class-floor balancing	Done	Underrepresented classes in the active augmented split were raised to a minimum of 300 images
Conservative external image additions	Done	Human-audited frost and drought additions were imported into train and propagated into train_aug
ResNet retrain on the rebuilt current split	Done	A fresh v4 checkpoint was trained on the rebuilt train_aug root and calibrated on its internal validation split
Latest ResNet checkpoint evaluation on active holdout	Done	The v4 checkpoint was evaluated on the active 70-image balanced holdout
Runtime promotion with rollback path	Done	The runtime app defaults to the current v4 checkpoint and keeps an explicit override path
p1/p2 runtime logging	Done	JSONL logging added for downstream calibration analysis
Tight dual-report generation pilot	Done	Focused 20-image pilot completed with 36 generated reports
Full stage-2 regeneration under tight policy	Pending	The current stage-2 JSON remains stale relative to the augmented training root
MiniGPT retrain on regenerated stage-2	Pending	Current checkpoint predates the holdout carve and dual-report policy
Clean holdout explanation evaluation	Pending	No post-regeneration holdout explanation metric exists yet
RF-DETR on/off ablation on clean holdout	Pending	Planned but not executed

A.2 Classifier Training and Calibration Details

The deployed ResNet-50 checkpoint is trained on the rebuilt `train_aug` root described in §3 with standard ImageNet normalization, moderate train-time augmentation, and a 256-pixel evaluation crop. The post-hoc temperature parameter $T = 0.7513$ in Eq. 1 is fit by minimizing negative log-likelihood on the internal validation split [6]. Reliability bins are constructed on a ten-bin equal-width partition of $[0, 1]$, following the protocol used in the calibration literature [11].

A.3 Grounding-Detector Training Configuration

The RF-DETR Small grounding detector is fine-tuned from a publicly released checkpoint with the following configuration:

- six-class plant-part vocabulary (flower, fruit, leaf, root, soil, stem);
- COCO-format dataset with 753 train images / 19,487 train annotations and 322 validation images / 8,582 validation annotations;
- 100 training epochs at resolution 560;
- batch size 8 with gradient accumulation factor 2 (effective batch 16);
- encoder learning rate 1.5×10^{-4} , decoder learning rate 1×10^{-4} , weight decay 1×10^{-4} ;
- cosine learning-rate schedule with five warmup epochs and minimum learning-rate factor 0.01;
- drop-path 0.1;
- exponential moving average enabled;
- seed 42;
- per-class runtime thresholds θ_p tuned per part as listed in §4.

A.4 Report-Writer Training Configuration

The active MiniGPT-v2 training configuration is summarized below. These values are repository-current settings; we reiterate that the deployed checkpoint predates the constrained stage-2 redesign and the holdout carve, and that retraining on the regenerated targets is identified as pending in Table 12.

- image size 448 for both train and evaluation processors;
- maximum text length of 500 tokens;
- LoRA rank $r = 64$, $\alpha = 128$, dropout 0.05;
- initial learning rate 1×10^{-5} , minimum 5×10^{-6} , warmup 1×10^{-6} ;
- 20 epochs with linear warmup and cosine decay;
- per-GPU batch size 2 with 8 gradient-accumulation steps;
- 10% auto-validation split;
- seed 42.

A.5 Runtime Decision Rule

The deployed runtime applies threshold-only acceptance under Eq. 3 with the per-class thresholds listed in Eq. 4, the default threshold of 0.55 for any unmapped class, and a post-prompt-assembly low-confidence guard that demotes accepted predictions whose top-1 probability falls below 0.50. The margin threshold of 0.08 is recorded to the structured decision log but is not currently activated as a runtime demotion rule; its operating point will be calibrated from production decision logs.

A.6 Scope of the Ambiguity Policy

The constrained dual-report policy of §4.6 is a dataset-construction policy. It does not at present imply a user-facing behavior in which the deployed report writer presents alternatives to end users. A user-facing alternative-presentation feature would require both a retrained MiniGPT-v2 checkpoint on regenerated targets and a clean holdout explanation evaluation, both of which are listed as pending in Table 12.

A.7 Per-Image Misclassifications on the Holdout

For qualitative inspection, Table 13 lists every misclassified holdout image for the deployed v4 checkpoint together with the calibrated top-1 probability p_1 assigned by the classifier. Eight of the 70 holdout images are misclassified; of these, five would be demoted to **unknown** by the deployed acceptance gate (Table 6), and the remaining three would be accepted. The accepted misclassifications are concentrated on the *overwatered* class, where the classifier transfers probability to *drought* or *frost* on examples whose visible symptoms are shared between drought and waterlogging.

Table 13: Misclassified holdout images for the deployed v4 checkpoint. p_1 is the calibrated top-1 probability assigned by the classifier; images with p_1 below the per-class acceptance threshold τ_c (Eq. 4) or below the 0.50 post-prompt-assembly floor are demoted to **unknown** by the deployed runtime gate.

Image	Ground truth	Predicted
drought_2.jpg	Drought	Gray mold
drought_23.jpg	Drought	Gray mold
GrayMold0144.jpg	Gray mold	Healthy
actzlp00vipa1.jpg	Overwatered	Frost
droopy-strawberry-just-got-my-dream-garden-plant-v0-dfb4nwahjv3d1.jpg	Overwatered	Drought
overwatered_2.jpg	Overwatered	Drought
WhiteMold0040.jpg	White mold	Gray mold
WhiteMold0045.jpg	White mold	Healthy

Misclassified holdout images for the deployed v4 checkpoint

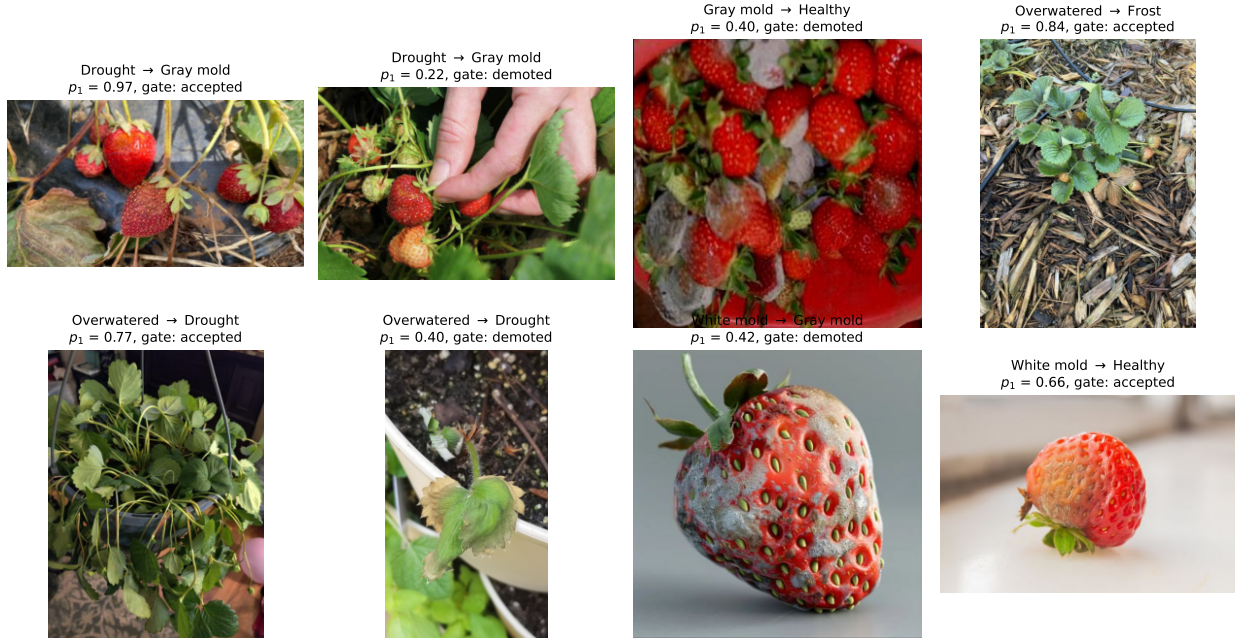


Figure 10: The eight misclassified holdout images for the deployed v4 checkpoint, with ground-truth $\rightarrow$ predicted labels, the calibrated top-1 probability p_1 , and the deployed gate decision (“demoted” images are sent to **unknown** at runtime; “accepted” images are forwarded to the report writer).

A.8 Reproducibility Manifest

The artifacts referenced in the main text are pinned as follows.

Source revision. Repository commit hash 6eaa403e021261f31f50e99a70c5b4a17944b497.

Model weights (SHA-256).

- ResNet-50 v4 classifier (plant_diagnostic/models/resnet_strawberry_v4_release10holdout.pth): 3d41c944c46b3f0bf5f356eb5e85df7f33afa69f2b121255421afc5558a569a2.
- RF-DETR Small detector (rfdetr_parts/output/checkpoint_best_total.pth): 0274206eb1f851835193a1c30fc455d8cf2edaf98b5fb26dff16d785b45a431a.

Evaluation artifacts. The classifier evaluation JSON is evaluation/holdout/resnet_v4_release10holdout_f. The dual-report pilot summary and per-record audit are evaluation/stage2_pilot/dual_report_focus20_tight, dual_report_focus20_tight_summary.json and ..._audit.jsonl.

Random seed. Both training pipelines fix seed=42. The bootstrap confidence intervals reported in §5.1 use a percentile bootstrap with 10,000 resamples and seed 42.

Build pipeline. All figures and tables in the manuscript are generated by the following scripts operating on the artifacts above:

- `paper/scripts/build_paper_assets.py` – pipeline figure, dataset counts, classifier confusion matrix, accuracy/calibration figure, dual-report pilot summary;
- `paper/scripts/compute_extra_metrics.py` – bootstrap confidence intervals, acceptance-gate replay, per-image misclassification table;
- `paper/scripts/compute_threshold_sweep.py` – threshold sensitivity sweep, risk-coverage curve, selective-accuracy table;
- `paper/scripts/compute_mcnemar.py` – McNemar paired test between v3 and v4 holdout predictions;
- `paper/scripts/render_misclass_grid.py` – per-image misclassification grid;
- `paper/scripts/run_rfdetr_on_holdout.py` – detector inference on the disease holdout, per-class firing-rate table and heatmap, latency summary (run inside the `rfdetr` conda env).

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